Installation and Commissioning Instructions

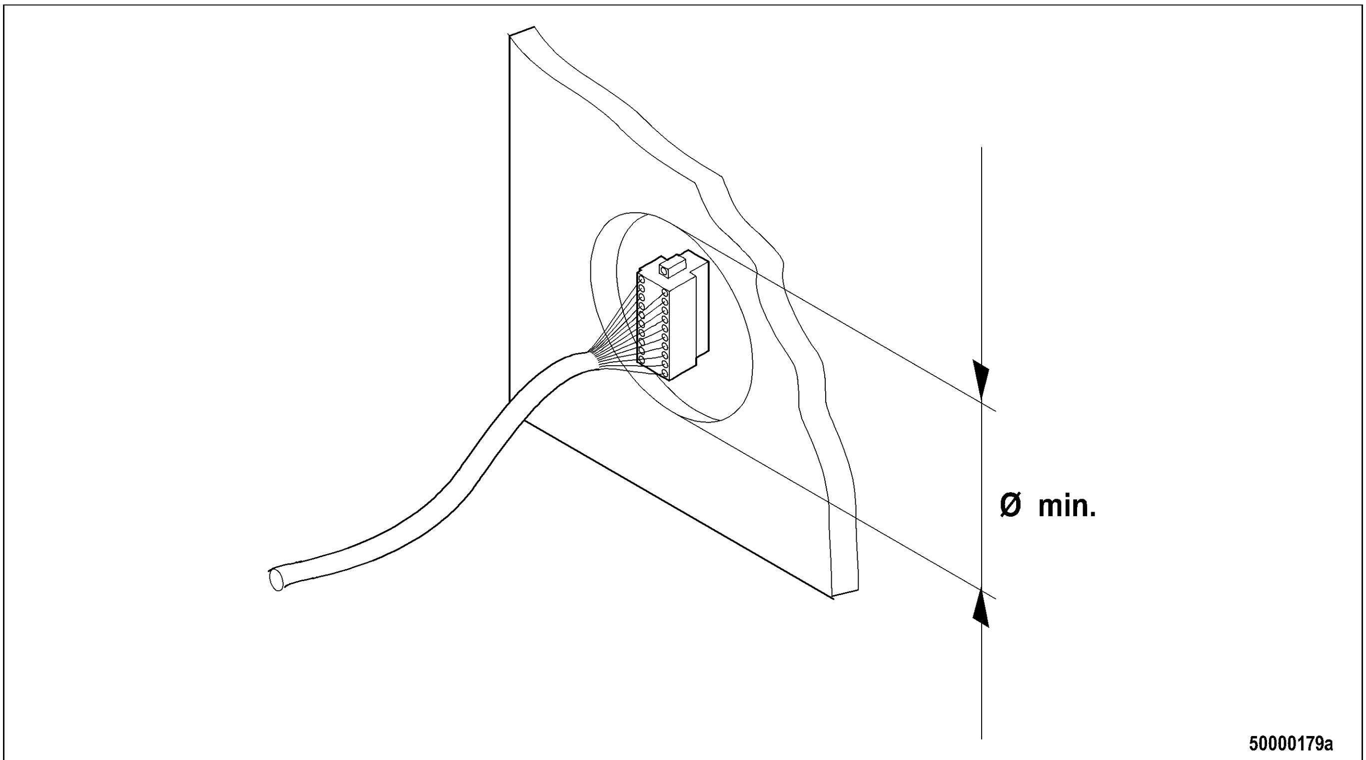
BlueLine Waterjet
Series 4000
ECS-5, MCS-5, RCS-5

10 Installation

10.1 Preparatory Work

10.1.1 Cable routes and openings between all installation locations – Check

Defining cable routes



1. Define the cable route between engine room and main control stand 1.
2. Define the cable route between main control stand 1 and main control stand 2 (if applicable).
3. Define the cable route between main control stand 1 and slave control stand 1 (if applicable).
4. Define the cable route between main control stand 1 and slave control stand 2 (if applicable).
5. Define the cable routes between the control units and the positions in the console where the MTU BlueLine devices are to be located.

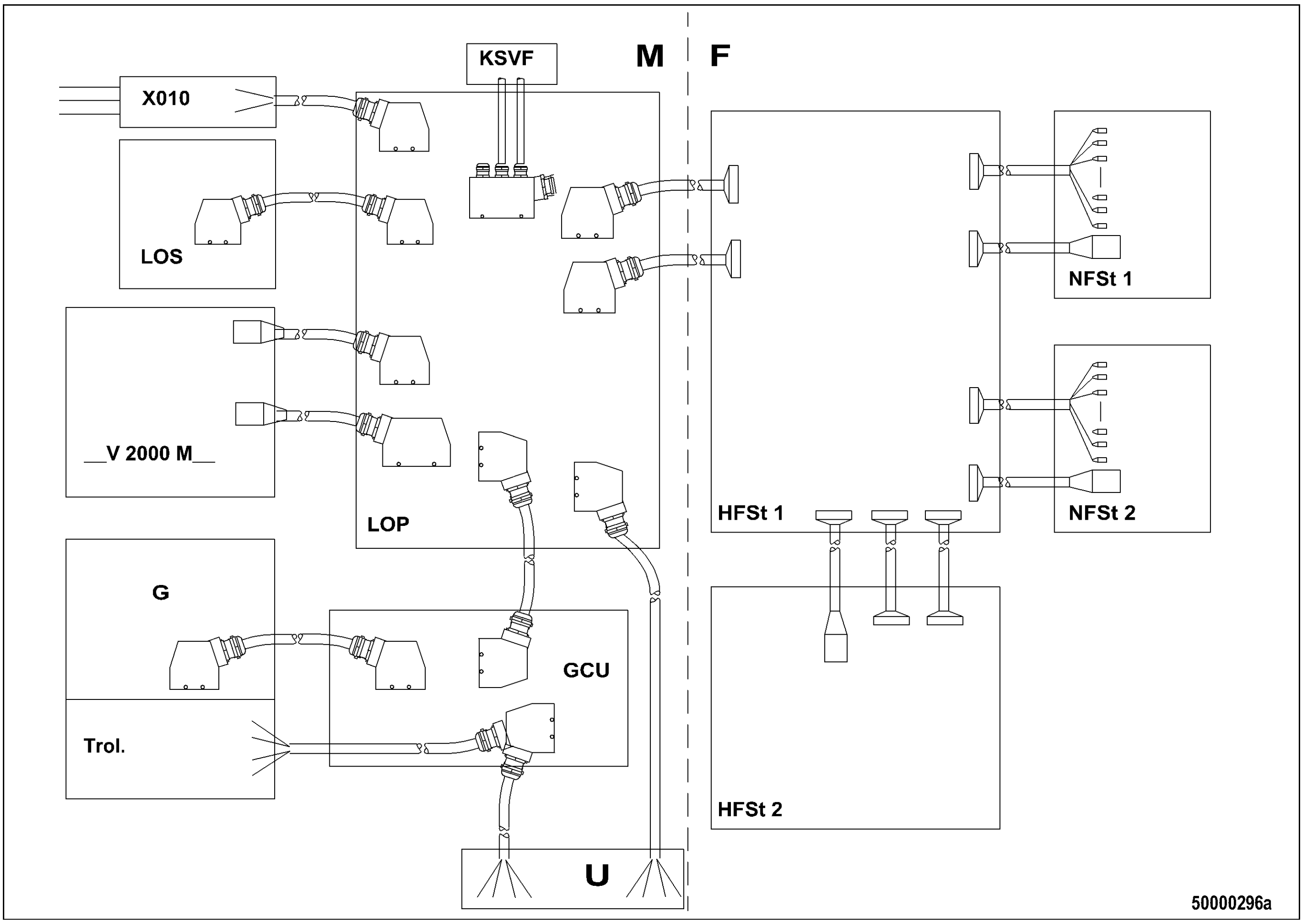
Checking cable routes

1. Check that the minimum opening diameters (see figure) through one or more partition walls over the course of these routes are no smaller than the dimensions specified below.
   1.1. Between LOS / LOP and machinery or engine room: min. 200 mm
   1.2. Between machinery or engine room and main control stand 1 (PIM 4): min. 100 mm
   1.3. Between main control stand 1 (PIM 4) and main control stand 1 (console front panel): min. 60 mm
   1.4. Between main control stand 1 (PIM 4) and slave control stand 1: min. 60 mm
   1.5. Between main control stand 1 (PIM 4) and slave control stand 2: min. 60 mm
   1.6. Between main control stand 1 (PIM 4) and main control stand 2 (PIM 4): min. 100 mm
   1.7. Between main control stand 2 (PIM 4) and main control stand 2 (console front panel): min. 60 mm
2. Change the cable route (check new cable lengths!) or increase the size of the openings if these specifications are not met.

10.1.2 Cables between installation locations – Routing

Preconditions

- Cable openings exist.
- Devices are not yet installed.



Cables between installation locations

Designation	Description	Comments
M	Engine room	Installation location for LOP, LOS (optional), GCU, X010
F	Control stands	Installation location for all control units, controls and indicators
__V 2000 M__	Engine 8/12/16 V 2000 M90/91	
LOS	Local Operating Station	Optional
LOP	Local Operating Panel	
GCU	Gear Control Unit	GCU 2 or GCU 3
G	Gearbox	
Trol.	Trolling device on gearbox	

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Designation	Description	Comments
X010	Auxiliary terminal box	
HFSt1	Main control stand 1	Always included, two control units
HFSt2	Main control stand 2	Optional, one control unit
NFSt1	Slave control stand 1	Optional
NFSt2	Slave control stand 2	Optional
U	Power supply	

1. Route the two cables between Engine Control Unit ECU (installed on the top of the engine) and Local Operating Panel LOP.
2. Route the cable between Local Operating Panel LOP and Local Operating Station LOS (only when optional LOS is included).
3. Route the cable between Local Operating Panel LOP and the Gear Control Unit.
4. Route the cable from terminal box X010 to Local Operating Panel LOP.
5. Route the pre-wired cable W014 on the generator from the engine to terminal box X010.
6. Route the pre-wired cable W014.1 from terminal box X010 to the starter.
7. Route the pre-wired cable W014.2 from terminal box X010 to the barring gear limit switch.
8. Route the cable from Gear Control Unit GCU to the gearbox.
9. GCU 3 only (Trolling option): Route the cable from Gear Control Unit GCU to the trolling device and to the starter batteries.
10. Route the cable from the Local Operating Panel to the “Water in fuel prefilter" sensor (to both sensors if applicable).
11. Route the cable between Local Operating Panel LOP and batteries.
12. Route the two cables between the engine room (Local Operating Panel LOP) and the control unit at main control stand 1.
13. Route the two cables between main control stand 1 and main control stand 2 and the cable for remote control (only if main control stand 2 is included).
14. Route the cable between main control stand 1 and slave control stand 1 and the cable for remote control (only if slave control stand 1 is included).
15. Route the cable between main control stand 1 and slave control stand 2 and the cable for remote control (only if slave control stand 2 is included).
16. All cables must be appropriately secured (e.g. cable clamps, cable ties) at suitable points on the vessel. Ensure that cables are routed neatly! Cover any sharp edges over which cables are routed (edge protectors, anti-kink protectors for cables).

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10.2 Mechanical Installation

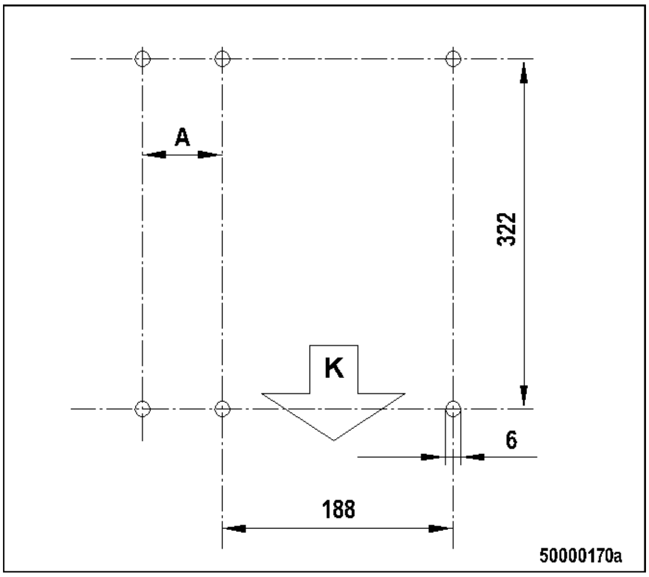
10.2.1 PIM 4 module housing – Installation

Preconditions

- Installation location is protected against moisture (control units have an ingress protection rating of IP 23).
- Installation location allows access to devices from the front after opening a cover, flap etc. (display visible).
- Installation location offers adequate space from below.

Installing housing

1. Mark out four mounting bores for the control unit on the wall inside a housing or console in accordance with the drilling pattern. Tolerance: ± 0.2 mm.
2. Ensure that adequate space is left to housing walls and floor:
   - Space between the two bores at the bottom and any bordering wall or floor: At least 130 mm
   - Distance A when several control units are mounted next to each other: At least 38 mm
3. Drill bores with a diameter of 6 mm.
4. Align the control unit straight on the wall and insert bolts through the mounting brackets and the bores in the wall.
5. Fit a plain washer, a spring washer and a nut on each of the four bolts on the other side; tighten by hand only.
6. Tighten the bolts with a suitable tool ensuring that the control unit is aligned horizontally.



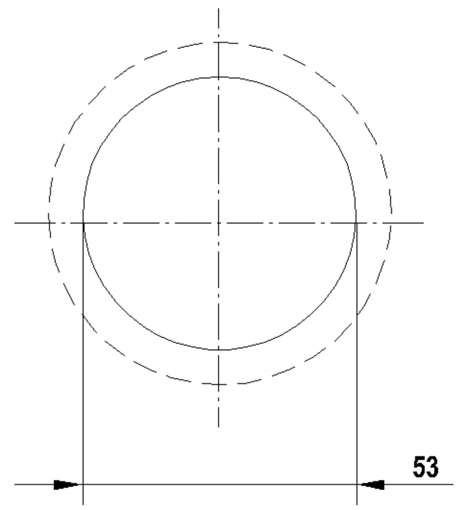
10.2.2 OceanLine analog display instruments – Installation

Preconditions

- Level contact area (broken line in figure below) of 120 mm and 62 mm in diameter is available.
- Minimum space between the center of the speed display instrument and the center of any other measuring instrument of at least 95 mm is provided.
- Minimum space between the center of any two small measuring instruments of at least 65 mm is provided.

Making the bore

1. Mark out the bore for the instrument concerned in accordance with the bore patterns (outside diameter of the bezel ring on a speed display instrument is 105 mm, outside diameter of the bezel ring on measuring instruments is 62.5 mm). The tolerance for the diameter of the mounting bores is ± 0.5 mm.



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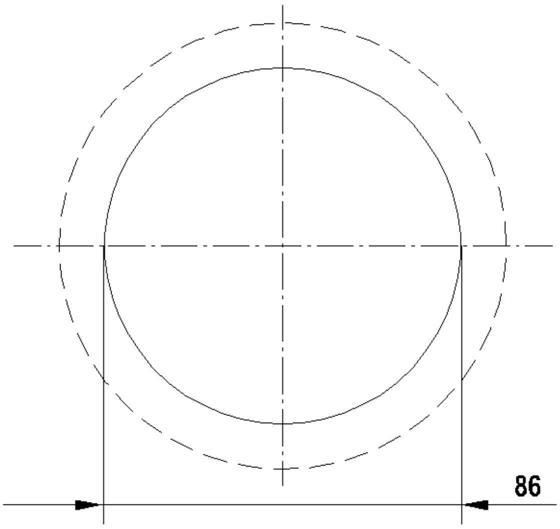
2. Make the bores with a suitable circular cutter.



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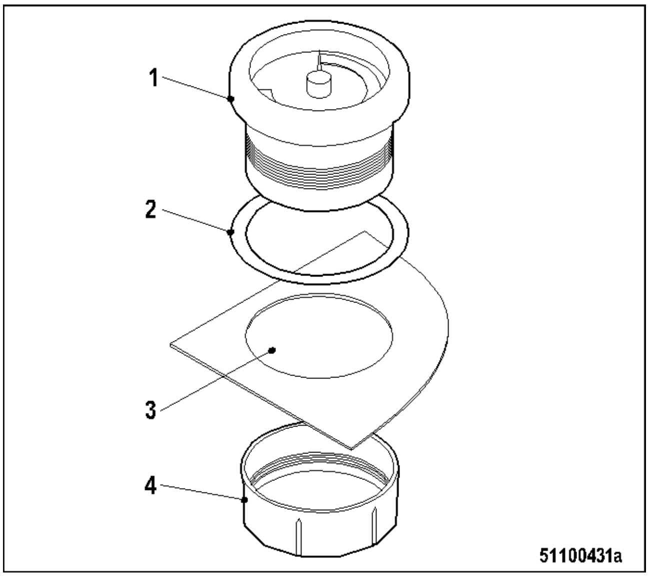
Installing the instrument

1. Unscrew the ring (4) of display instrument (1).
2. Insert instrument (1) with seal (2) into the installation opening (3) from the top.
3. Fit the ring (4) on the back of the instrument.
   3.1. If the thread bites, screw the ring on a few turns; ensure that the ring is not cross-threaded.
   3.2. If the thread fails to bite (e.g. due to console front plate thickness > 5 mm), turn the ring over and fit.
   3.3. Screw the ring on a few turns; ensure that the ring is not cross-threaded.
4. Adjust instrument (1) from the top (scale / inscription horizontal) and hold it in place.
5. Tighten ring (4) by hand.



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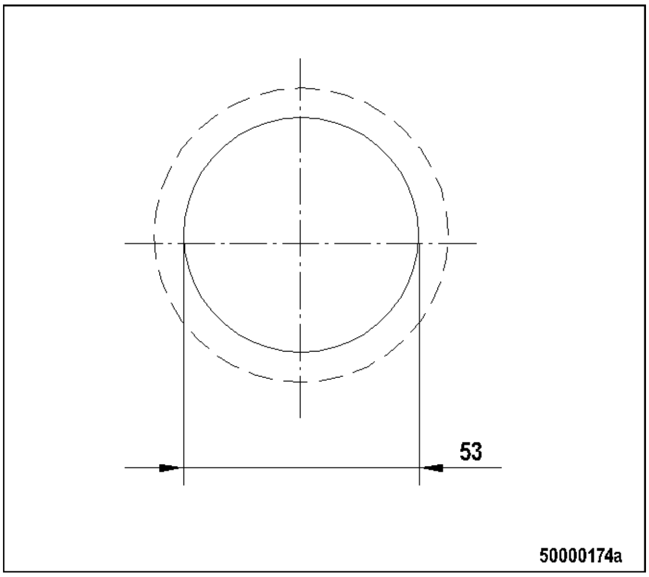
10.2.3 VDO horn – Installation

Preconditions

- The seating surface (broken line in figure) of 63 mm in diameter must be level to ensure tight sealing between the horn and the surface.

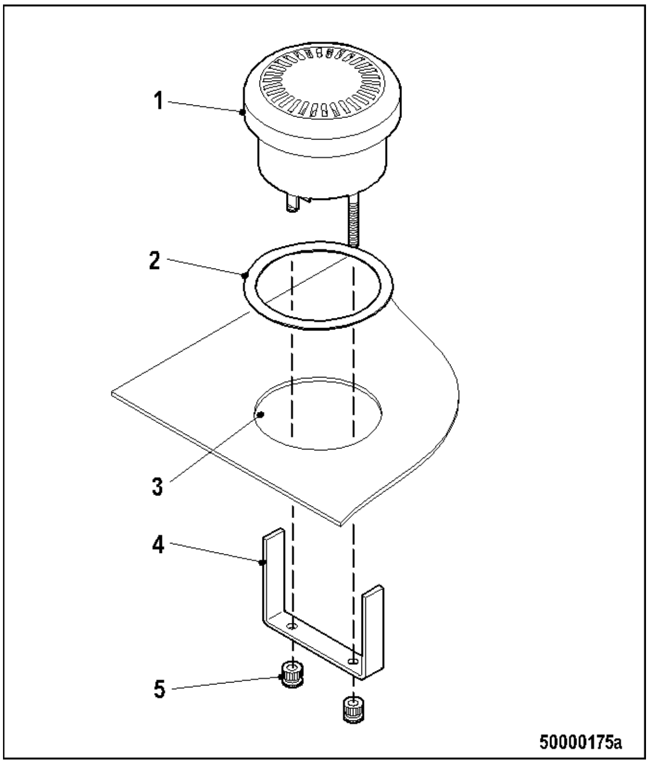
Making bore

1. Mark out the bore for the horn in accordance with the bore pattern. Tolerance for the diameter of the bore is ± 0.5 mm.
2. Make the bores with a suitable circular cutter.



Installing horn

1. Unscrew the horn retaining bracket.
2. Insert the horn (1) and rubber seal (2) into the appropriate installation opening (3) from the top.
3. Fit the retaining bracket (4) onto the back of the horn. Screw on the two knurled-head screws (5) a few turns.
4. Adjust the horn (1) from the top (inscription horizontal) and hold it in place.
5. Tighten both knurled-head screws (5) firmly by hand.



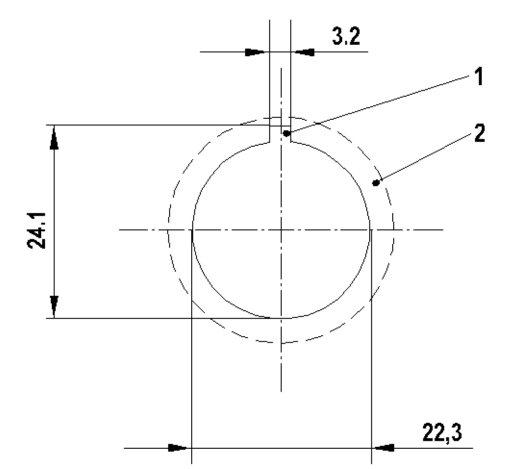
10.2.4 Indicator lamp – Installation

Preconditions

- The seating surface (broken line in figure) of 28 mm in diameter for each indicator lamp must be level to ensure tight sealing between the indicator lamp and the surface.
- The minimum horizontal distance between the centers of any two illuminated pushbuttons / indicator lamps is at least 31 mm.

Making bores

1. Mark out bores for all the indicator lamps at their intended locations in accordance with the bore pattern. Tolerance for all dimensions and distances between bores: + 0.3 mm.
2. Make 22 mm bores with a circular cutter.
3. Make an anti-rotation notch (1) in each bore (e.g. with a key file). It must be made on the side of the bore facing away from the operator.



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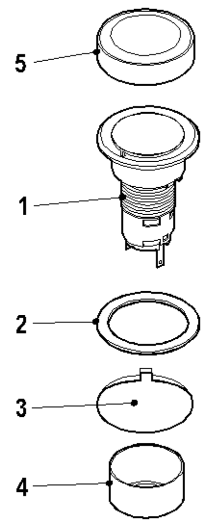
Installing housing

1. Insert the housing (1) through the bore (3) from the top.
2. Insert the right indicator lamp (with the appropriate inscription). Ensure that the indicators lamps are installed in the right order.
   2.1. Enclosed control stands: The seal (2) must be fitted between the console surface and the edge of the housing.
   2.2. Exposed control stands: Fit the sealing cap (5) over the edge of the indicator lamp (1). Seal (2) is not required.
3. Screw on the central nut (4) from below by hand.
4. Tighten the central nut (4) taking care not to overtighten it.



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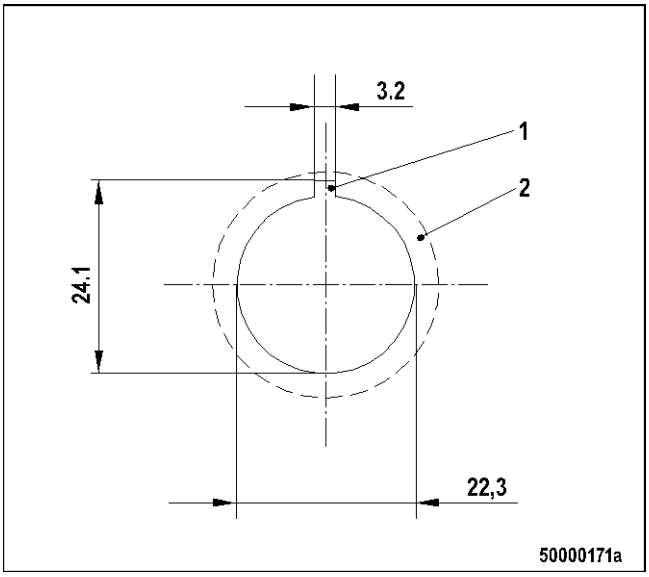
10.2.5 Illuminated pushbutton / key switch – Installation

Preconditions

- The seating surface (2, broken line in figure) of 28 mm in diameter for each illuminated pushbutton must be level to ensure tight sealing between the illuminated pushbutton and the surface.
- The minimum horizontal distance between the centers of any two illuminated pushbuttons / indicator lamps is at least 31 mm.

Making bores

1. Mark out bores for all the illuminated pushbuttons at their intended locations in accordance with the bore pattern. Tolerance for all dimensions and the distances between bores: + 0.3 mm.
2. Make 22 mm bores with a circular cutter.
3. Make an anti-rotation notch (1) in each bore (e.g. with a key file). It must be made on the side of the bore facing away from the operator.



Installing actuator

1. Insert the right actuator (with the appropriate inscription). Ensure that the actuators are installed in the right order.
   1.1. Enclosed control stands: The seal (2) must be fitted between the console surface and the edge of the housing.
   1.2. Exposed control stands: Fit the sealing cap (5) over the edge of the indicator lamp (1). Seal (2) is not required.
2. Screw on the central nut (4) from below by hand.
3. Tighten the central nut (4) taking care not to overtighten it.
4. Clip on the corresponding contact unit (5) with the LED inserted.